

Child-Proof Door Latch
 MECH 210: Engineering Design — 3D Modeling and Printing
 Colorado State University, Department of Mechanical Engineering
 Jack Friesen, Carlos Rivera, Logan Tschakert, Quentin Mays
 Lab 4 Group 1
 Hunter Brumblay
 Dr. Bonnie Roberts
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Name of Section	Individual Contribution	Editor(s)
Cover Page	Jack Friesen	Carlos Rivera
Introduction and Problem Definition	All	All
Design Requirements and Criteria	All	All
Concept Development and Design Evolution	All	All
Final Design Description	All	All
Design for X (DfX) Considerations	All	All
Manufacturing and Assembly Process	All	All
Testing and Evaluation	Jack Friesen, Carlos Rivera	None
Design Iteration and Improvement	All	All
Final Assessment and Discussion	All	All

Project Contributions	Individual Contributions
SolidWorks Modeling	Jack Friesen, Logan Tschakert
3D printing	Jack Friesen, Carlos Rivera, Quentin Mays
Assembly	Jack Friesen
Testing	Jack Friesen, Carlos Rivera

Introduction and Problem definition

In many households, cabinets are used to store items such as cleaning chemicals, sharp objects or medications that can pose a safety risk to children. Although commercial child safety locks already exist, they often require complex installation or are inconvenient for daily use by adults. This creates an issue in which a low cost, and easy to use solution can be used to maintain reliable safety performance.

The objective of this project was to design a cabinet locking mechanism that balances both child resistance with adult usability while also being able to be manufactured by the use of FDM 3D printing. The device needed to function passively (Automatic lock) while requiring intentional coordinated actions to prevent children from unlocking it accidentally.

The primary users of this device are households with young children requiring reliable protection from cabinet hazards. Key user needs include dependable locking performance under significant pulling force, straightforward operation for adults, and an overall low-cost design that remains accessible to a wide range of households. The design is further constrained by the material strength limitations inherent to FDM 3D printing, as well as the need to accommodate varying cabinet dimensions found in typical home environments.

Design Requirements and Criteria

The design requirements for the Child-Proof Door Latch were developed based on the need to balance safety, usability, and manufacturing. Each requirement directly influenced design decisions and was paired with criteria that need to be followed to evaluate its performance during testing.

Requirement	Type	Design Decision	Evaluation Criteria
Automatic locking when cabinet closes	Functional	One way latch with a spring based trigger	Locks consistently without user input.
Prevent opening under unintentional force	Functional	Spring resistance	Withstands applied pulling force without disengaging
Child resistance	Safety	Internal trigger placement requiring intentional pull	Can be opened by an adult but difficult for a child
Durability under repeated use	Performance	Use of spring for consistent force	No cracking or failure after repeated cycles(we still have to test this)

Compatibility with different cabinets	Constraint	Compact design with mounting holes for standard cabinets	Fits and functions within typical cabinet dimensions
Low manufacturing cost	Constraint	3D printed design with minimal material and part count	Low print time and material usage
Ease of manufacturing	Manufacturing	Optimized orientation, simple geometry	Prints successfully without supports
Ease of assembly	Assembly	Three main components with simple mounting assembly	Can be assembled quickly with minimal issues

Concept Development and Design Evolution

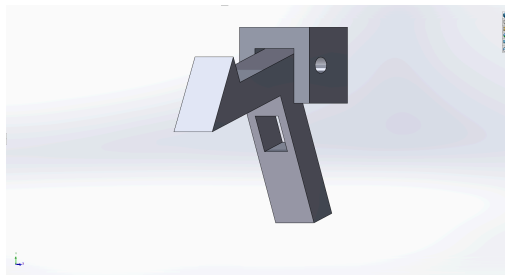


Figure 1: Initial Assembly

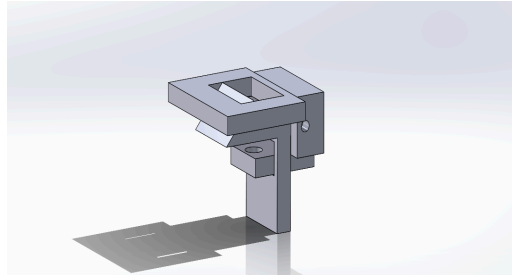


Figure 3: Third Iteration

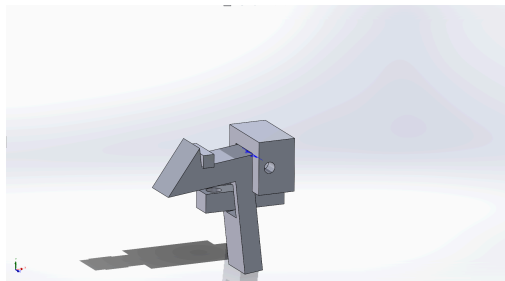


Figure 2: Second Iteration

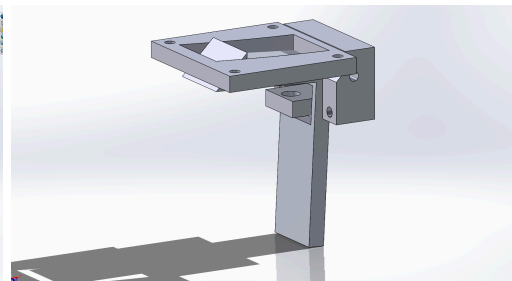


Figure 4: Latest Iteration

The initial concept for the cabinet safety lock remained consistent throughout the design process focusing on a spring loaded latch mechanism that allows for automatic locking and controlled release. Instead of changing the entire project the design changed throughout a series of iterative improvements aimed at enhancing efficiency, and functionality.

Figure 1 (Initial Assembly) shows the early version of the design which established basic iteration between the holder and the latch. At this point the geometry was very simple and primarily focused on getting an idea of the system and lacked functionality.

Figure 2 (second Iteration) introduced changes to improve the interaction between the trigger and holder. Adjustments were made to the contact of the surface to create smoother engagement and reduce the chance of misalignment during use of the mechanism.

Figure 3 (Third Iteration) focused on improving structural stability and use of the mechanism. Additional features were added to better support the trigger mechanism and improve how components fit together. The geometry of the housing was improved to provide a more consistent alignment and reduce unwanted movements to prevent the latch from opening on its own.

Figure 4 (Latest iteration) represents the final design, where all components are fully integrated and optimized. The latch interface, trigger motion, and mounting were adjusted to ensure reliable use of the system and easy assembly. The overall geometry was also improved to better fit 3D constraints such as improving consistency.

Although the concept remained the same these iterations in our project resulted in a more reliable, manufacturable and user friendly final design.

Final Design Description

Overview of the Final Design

The final design is a 3D printed, spring loaded cabinet safety lock that uses a trigger arm motion guided by the geometry of the housing, while a discrete off-the-shelf spring provides the restoring force required for operation to prevent unintended access to cabinets. The device is designed to automatically engage when the cabinet is closed and requires a trigger action to release, making it suitable for child safety applications while remaining convenient for adult users.

The design focuses on simplicity, low cost, and ease of assembly by minimizing part count and utilizing a latching system.

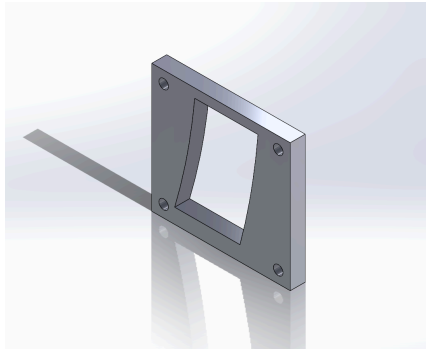
Functional Operation of the Device

The mechanism works based on a one way latch interaction between the trigger and the insert plate. When the cabinet door is closed, the angled face of the latch forces the trigger inward, temporarily deflecting the flexure. Once the latch passes the trigger the flexure returns the trigger to its original position locking the cabinet. To open the cabinet, the user needs to manually pull the trigger, overcoming the force and disengaging the flexure from the insert latch. This motion requires intentional input thus preventing accidental or unintended releases. Although a simple design it ensures both passive locking as well as controlled unlocking

Major components and Their roles.

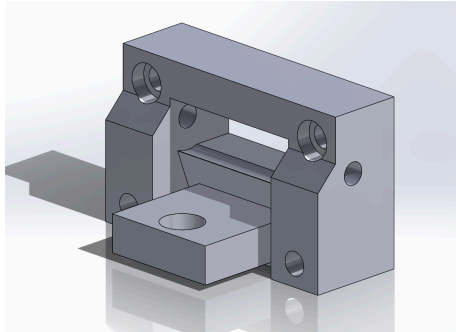
The final design consists of three 3D printed components:

1. Insert Latch Plate



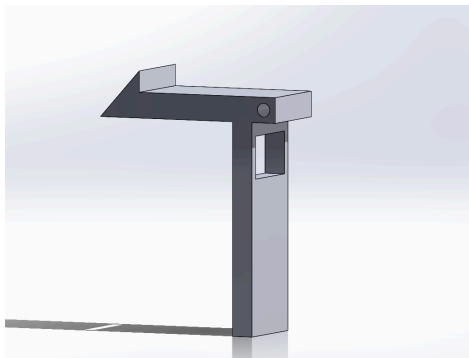
The insert plate is mounted to the roof of the cabinet frame and serves as the primary locking interface. It includes an angled surface that allows the trigger to slide over it during closing and a flat surface that prevents reverse motion when engaged.

2. Holder



The holder housing is mounted to the cabinet door and acts as the support for the trigger mechanism. It aligns the trigger arm with the latch plate and provides mounting screw holes for installation. The housing also constrains the motion of the trigger arm ensuring constant engagement with the latch,

3. Trigger arm



The trigger arm is the primary moving component of the mechanism and pivots within the holder housing. Its motion is guided by the geometry of the housing, ensuring consistent interaction

with the latch. A spring is incorporated into the system to provide restoring force, allowing the trigger to return to its engaged position after deflection.

3D Printed Components and Design Considerations

All three main components, The insert latch plate, The holder, and trigger arm were designed to be 3D printed using Fused Deposition Modeling (FDM).

These parts were designed with these considerations:

- Reduce support material
- Ensure right wall thickness for structural integrity
- Designing the Trigger arm to align with print layers to improve durability
- Provide sufficient tolerance between moving part to account for printer variability

Design for X (DfX) Considerations

Design For Additive Manufacturing (DfAM)

The Design of the cabinet safety lock was developed with the consideration of the printing constraints of a FDM 3D printer, including print orientation, support material, and dimensional tolerances. To ensure successful print, the components were oriented to maximize contact with the print bed, improving stability and reducing the risk of print failure.

Due to the geometry of the trigger arm and some internal features of the holder housing some overhanging regions required support materials during printing. While supports increase material usage and post-processing time, they were necessary to maintain dimensional accuracy and ensure proper functionality of the mechanism.

Tolerances between moving components, such as the trigger and holder mount, were adjusted to account for the variability of a 3D printer. Some clearances were introduced to allow for a smooth motion without requiring extensive post processing.

The final design incorporates an off the shelf spring rather than a fully printed spring mechanism. This decision was made because common 3D printers are not suited for the creation of springs. The use of a manufactured spring provides more consistent force and also has a better quality that leads to an improved durability during repeated use.

Design for Assembly (DfA)

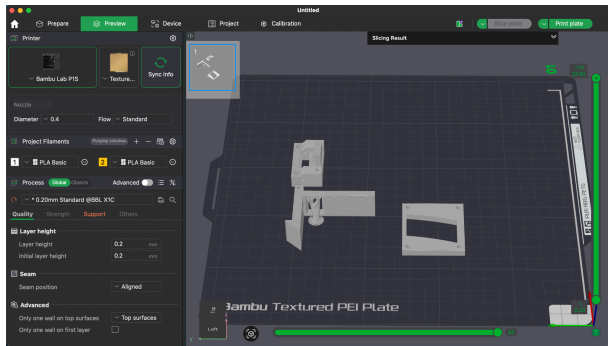
The Design was simplified to include three main 3D printed components and a single spring reducing the overall assembly complexity while maintaining functionality. Limiting the number of parts minimized potential failure points and simplified the assembly process

One of the main assembly challenges faced during the design was proper alignment between the latch plate and the trigger arm. Early iteration shows that small misalignments could lead to inconsistent engagement or failure to lock properly. To address this the geometry of the holder

was made to align the trigger arm properly with the insert latch with the use of screws to the cabinet.

Mounting features were designed to allow a straight forward installation with clear attachment points within the cabinet to reduce errors.

Manufacturing and Assembly Process



The components of the cabinet safety lock were manufactured using FDM 3D printing with PLA filament. A layer height of 0.16mm was selected to balance the surface quality and print time. The parts were oriented flat on the print bed to maximize the stability and ensure strong adhesion during printing. This orientation also helped improve dimensional accuracy for mounting surfaces and interfaces between different components.

Due to the geometry of the trigger arm and internal features of the holder, housing support material was required for specific regions to properly form overhanging structures. Multiple components were printed simultaneously in a single build to improve manufacturing efficiency and reduce total production time. Post processing included removal of the support material and minor clean ups to ensure smooth interaction between moving parts.



Testing and Evaluation

Requirement	Test	Result
Automatic locking when cabinet closes	When door closes, latch automatically locks	Pass; latch automatically locks

Prevent opening under unintentional force	Pull suddenly and with force on the handle	Pass; cabinet stays closed with sudden forces
Child resistance	8 lbs of force applied to the handle	Pass; over 8lbs of force was exerted over a long period of time and cabinet stayed closed
Durability under repeated use	Open and close cabinet for 5 minutes straight	Pass; over the course of 5 minutes, every single latch successfully locked
Compatibility with different cabinets	Install on multiple different cabinets	Fail; design only works for custom cabinet at this time
Low manufacturing cost	All custom parts cost less than \$1 total	Pass; 3d print is 18.29g. At 2 cents per gram, that is well below \$1total
Ease of manufacturing	Takes less than 1 hour to 3d print	Pass; print time is 1 hour exactly
Ease of assembly	Takes less than 5 minutes to install onto cabinet	Pass; assembly took under 5 minutes total

Overall, the cabinet door latch passed with flying colors. The largest issue it faced while testing was the fact that it was only able to be installed on the custom cabinet. The reason this occurred was because access to an actual cabinet was not possible at this time.

Design Iteration and Improvements

Design Refinement: A key improvement involved the trigger arm and latch plate interaction. Early iterations suffered from inconsistent alignment, causing unreliable engagement and excess friction. The trigger geometry and holder housing were refined to improve contact alignment, resulting in more reliable locking, smoother operation, and better durability.

Print Parameters: Components were sliced in Bambu Studio and printed on a Bambu Lab P1S using PLA Basic, 0.16 mm layer height, gyroid infill, and three-perimeter walls. Infill density was 25% for the latch plate, holder housing, and clips. Supports were enabled for the trigger arm overhang and the spring seat cavity, with interface layers used to improve surface finish and ease removal.

Assembly Process and Challenges: Assembly involved three steps: (1) inserting the trigger arm into the holder housing channel and seating the spring; (2) securing the holder to the cabinet door with M3 screws; and (3) attaching the latch plate to the cabinet frame at a height calibrated for full engagement at door close. The main challenge was spring retention during

step one, as the spring tended to unseat before the trigger arm was fully inserted. This was resolved by compressing the spring manually and inserting the trigger arm at a slight angle before rotating it into place. A future improvement would add a retaining tab to the housing to hold the spring during assembly.

Final Assessment and Discussion

The final design of the cabinet safety lock successfully meets its primary objectives of providing a low cost, reliable, and easy to use solution for preventing unintended access to cabinets. The mechanism achieves automatic locking upon closure and requires intentional user input to release, effectively balancing child resistance with adult usability.

One of the main strengths of the design is its simplicity. By limiting the system to 3 main components that are 3D printed and a singular spring the device maintains a low part count while still achieving functional reliability. The use of a spring provides constant force, improving durability and performance over repeated use.

However this design also has its limitations. The performance of the mechanism is sensitive to alignment between components, which may change depending on installation accuracy. The reliance on 3D printing also introduces variability in the tolerances and surface finish which can affect the smoothness of operation. Additionally, the need for support material during printing increases post processing time and material use. Increasing tolerances, using smarter support structures, and optimising print orientations would significantly improve a future design to overcome these limitations.

Overall the project demonstrates how iterative design, combined with consideration of manufacturing and assembly can lead to a practical and functional mechanism solution.